Computer Science GCE A Level 2024 - Paper 2

Duration: 2h30 | Cameroon GCE Board

General Certificate of Education Board - General Certificate of Education Examination. Computer Science 2 (0795), ADVANCED LEVEL, June 2024. Duration: two and a half hours.

Answer any SIX questions. All questions carry 17 marks each. For your guidance, the approximate mark for each part of a question is indicated in brackets. You are reminded of the necessity for good English and orderly presentation in your answers. In calculations, you are advised to show all the steps in your working, giving your answer at each stage.

Exercise 1 (17 marks)

i. Using two's complement, show how the following operations on the denary numbers could be performed in a 4-bit register:

(a) $5 - 4$.

(b) $-3 + 6$.

(4 marks)

ii. Express the AND gate and the OR gate in terms of NAND gates only.

(4 marks)

iii. (a) Explain how the width of the data bus **and** system clock speed affect the performance of a computer system. (3 marks)

(b) The table below shows six stages in the fetch-execute cycle in a Von Neumann computer. Put the stages into the correct sequence by writing the numbers 1 to 6 in the right-hand column. In your answer booklet, reproduce this table, replacing each description on the left with its corresponding letter (A to F). (3 marks)

Description of Stage	Sequence Number
A. The instruction is copied from the Memory Data Register (MDR) and placed in the Current Instruction Register (CIR).	
B. The instruction is executed.	
C. The instruction is decoded.	
D. The address contained in the Program Counter (PC) is copied to the Memory Address Register (MAR).	
E. The value in the Program Counter (PC) is incremented so that it points to the next instruction to be fetched.	
F. The instruction in the memory location found in the Memory Address Register (MAR) is copied into the Memory Data Register (MDR).	

iv. Perform a 2 place logical right shift on the binary number `11001011`. Convert the binary number obtained to decimal and state the effect of performing a 2 place logical right shift on the binary number. (3 marks)

Exercise 2 (17 marks)

i. Consider the following assignment statement written in a high-level language, where A is an array, = the assignment operator and array indexing starts with 0.

$$A[3] = A[2] + A[1]$$

(a) Explain what is meant by an *addressing mode*. What is absolute addressing? (2 marks)

(b) Using register and indexed addressing modes only, show how you would implement the above statement in assembly. Assume that the base address of array A is stored in a register R0, each array element occupies a single byte and other registers are named $R1, R2, \ldots, Rn$, for some positive integer n. Explain any other notation you use in your answer. (4 marks)

ii. A three-bit binary number given as ABC, where A is the most significant bit and C the least significant bit, appears on the input to a combinational logic circuit. The circuit has two outputs X and Y. When the input number, $(ABC)_2$, is divisible by two (10_2), output X is 1 and when the input number is divisible by three (11_2), output Y is 1. Otherwise, X and Y are 0.

(a) Explain what a combinational logic circuit is. (2 marks)

(b) Draw a truth table for this circuit with inputs A, B, C, and outputs X and Y. (4 marks)

(c) State the most simplified Boolean expression for output X. (1 mark)

iii. Bezea has taken lots of photos and the memory in her camera is full. She needs to transfer her photos to an external secondary storage device. Ignoring cost, describe **two** characteristics of secondary storage devices that Bezea should consider when choosing a device. (4 marks)

Exercise 3 (17 marks)

i. (a) How is the concept of *virtual memory* used in memory management? (2 marks)

(b) Explain how paging may be used to support virtual memory. (4 marks)

(c) State **two** primary benefits of virtual memory. (2 marks)

ii. The table below shows three processes P1, P2, and P3 with their corresponding arrival times and burst times. These processes are to be scheduled using a pre-emptive scheduling policy.

Process ID (PID)	Arrival time (AT)	Burst time (BT)	Completion time (CT)	Turnaround time (TT)	Waiting time (WT)
P1	0	7			
P2	1	3			
P3	3	4			

(a) What is meant by pre-emptive scheduling? (1 mark)

(b) Round-robin (RR) and shortest remaining time first (SRTF) are two pre-emptive scheduling algorithms. Explain when pre-emption takes place in each of RR and SRTF algorithms. (2 marks)

(c) Using SRTF scheduling, copy and complete the table above by filling in the completion time, turnaround time and waiting time for all three processes. *Replace column headings with the abbreviations in brackets.* (4 marks)

(d) Calculate the average turnaround time and average waiting time for these processes. (2 marks)

Exercise 4 (17 marks)

i. In relation to database design and management, define the following terms:

(a) Database management system (DBMS). (2 marks)

(b) Candidate key. (2 marks)

(c) Multivalued attribute. (2 marks)

(d) Normalisation. (2 marks)

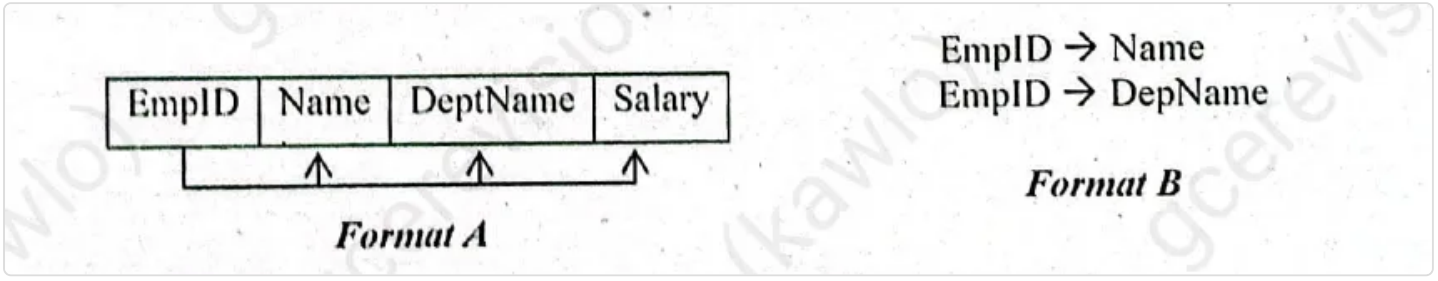
ii. Use the database below to answer the questions that follow.

StudentId	StudentName	CourseId	CourseName	Grade
001	Ngwa	CSC01, ICT01	Computer Sc., ICT	A, B
002	Abubakar	IT01	Technology	A
003	Wankah	NET01, CSC01	Network basics, Computer Sc.	B, C
004	Chifu	CST01	Computer technology	B

(a) With a suitable example, explain why the above table is NOT in First Normal Form (1NF). (2 marks)

(b) Convert the table above into one in First Normal Form (1NF). Write out only the fields and records that are changed. (2 marks)

iii. Answer the questions below using either of the formats A or B, to represent functional dependencies.



(a) Identify two functional dependencies from the table in (ii) above. (2 marks)

(b) Convert the table in (ii)(b) above into Second Normal Form (2NF). Represent your 2NF tables using the format: *Table_name(field1, field2, ...)* (3 marks)

Exercise 5 (17 marks)

i. A software developer retains the source code for a particular application and provides the application's user with the object code.

(a) Explain the difference between source code and object code. (2 marks)

(b) State **two** reasons why the application's user would be provided only with the object code. (2 marks)

ii. Describe **three** different types of errors that can occur in a computer program, giving a specific example of each. (6 marks)

iii. Consider the following array of colors to sort in alphabetical order.

```
[ Green | Blue | Yellow | Red | White ]
```

(a) For each pass in a bubble sort algorithm, use arrows to indicate how values in array cells are swapped, then write out the array cell values at the end of each pass. (4 marks)

(b) Show the array ranges during each stage of a binary search to find the color *Yellow* in the sorted array in (iii)(a) above. (Hint: The mid-position is the integer part of its calculation.) (3 marks)

Exercise 6 (17 marks)

i. (a) State what the computational complexity of an algorithm is and give an intuitive explanation of how it is evaluated. (2 marks)

(b) Using the big-O notation, give the tight upper bound for the algorithms given as follows. (4 marks)

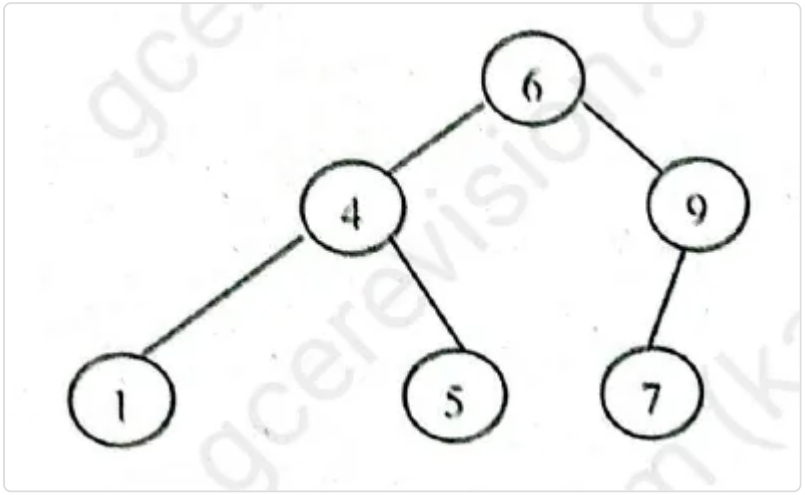
1. $n^4 - 5n + 4$

2. The following algorithm:

```
for i = 1 to n do
    for j = 1 to n do
        write(i, j)
```

ii. What is an intractable problem? Give an example. (2 marks)

iii. Consider the tree below and answer the questions that follow.



(a) Taking into account the node contents and their placement, how is this type of tree called? (1 mark)

(b) What type of tree traversal must be used if we must output the tree nodes in an ascending or descending order? Explain how it would be done. (3 marks)

iv. (a) Draw the binary tree that corresponds to the expression $a * b - c/d$. (3 marks)

(b) Write out the tree if it is output breadth-first. (2 marks)

Exercise 7 (17 marks)

i. An object-oriented program is being written to store details of and play media files that are stored on a computer. A class **MediaFile** has been created and two subclasses, **VideoFile** and **MusicFile** are to be developed. The classes VideoFile and MusicFile are types of MediaFile. In object-oriented programing,

(a) Give a diagrammatic representation of these classes. State and justify the object-oriented concept that best relates the classes. (3 marks)

(b) Give in pseudocode, a method (loosely, a procedure or function) to print the artist of a **Music_file** and another method by the same name to print the title of a **Video_file**. What main object-oriented programming language concept permits one to call this method with any media file object and correctly print the right values? Justify your answer. (4 marks)

(c) Briefly explain how one would exploit encapsulation in order to make sure methods in **MusicFile** object do not access or modify those in **VideoFile**. (3 marks)

ii. (a) For children, a tourist attraction has a daily charge of 5000 FCFA on weekdays and 7500 FCFA on weekends or bank holidays. Adults are charged 8000 FCFA on weekdays and 12000 FCFA on weekends and bank holidays. Write a pseudocode algorithm that allows the user to calculate the charge for a visitor. (4 marks)

(b) Explain in words how you could modify your program in (ii)a so that it recursively calls itself but also checks that the recursion terminates. (3 marks)

Exercise 8 (17 marks)

A college currently uses two separate *local area* networks (LANs), one for administration and the other for teaching.

i. Give three reasons why using a LAN to connect computers is well-advised, and one main reason why separate LANs should be used for administration and teaching. (3 marks)

ii. The administration network consists of 12 workstations and a dedicated file server. One of the workstations also acts as a print server. Explain why this configuration for workstations and printing could be preferable over connecting the printer directly to the LAN. (4 marks)

iii. *Bus*, *Ring* and *Star* are common topologies for LANs. Describe these topologies, using clearly labeled diagrams to illustrate your answer. (6 marks)

iv. The college wishes to connect the administration and teaching networks so that teaching staff can have access to student records. A bridge is proposed to make the connection.

(a) Briefly explain why a bridge could be inadequate as a possible solution. (2 marks)

(b) State **two** functions provided by a gateway that would make it preferable over a bridge or are not provided by a bridge. (2 marks)

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